Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, November 10, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Java Coding Toolsets

As you know, the *Java* coding environment provided by BHCC for this course is the "**Grader Than**" web-based toolset. This provides our students the capability of completing course assignments using a web browser on their own computer. In this course, some students have chosen to use some other *Java* coding toolset.
If the environment you are using supports **JavaFX**, then you may continue to do your work that way. However, if your alternative tools do not support **JavaFX**, then you will need to use "**Grader Than**".

The GraderThan Environment

If you are using **GraderThan** for the first time, follow the instructions in the

Lab01a_GraderThan_Intro_20250908

document in the **Lab Exercises** topic on ***Moodle***.

Setting up a JavaFX Project using GraderThan

On your local PC, download the ZIP file associated with **Lab14a_Begin_JavaFX…** on *Moodle*. Extract the **Lab14a.java** "starter code" file from the ZIP drive. Make a note of the **folder on your PC** where you have saved this file.

1. Login to the **GraderThan** environment:

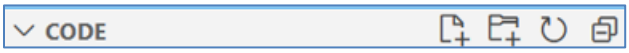
   If you have already been using **GraderThan** for your CSC239 Java Programs, and you have been placing your **Lab** solutions in a particular directory that you have created, **then** you may navigate to that directory and skip ahead to the **Create the new Java Project** section of this document. **Otherwise**, continue following the instructions below:

2. Notice the "menu" button (3 horizontal lines) near the upper-left corner of the display: 
3. Left-click the "menu" button and select **File→OpenFolder…**
4. Navigate to the **/home/developer/Documents/code/** directory and click "**OK**".
5. The "Explorer" section of the screen will look something like



6. Click on 

to "expand" it and show several buttons:



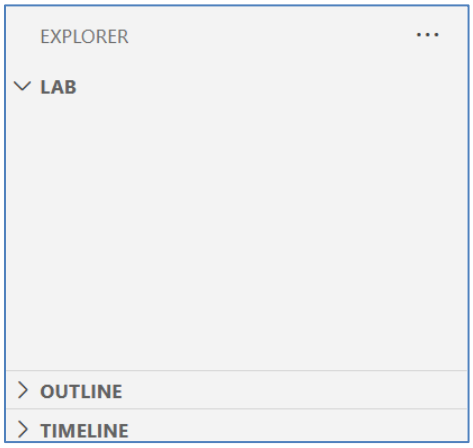
Click on the  button to create a new sub-directory: **Lab**

7. Use the **File→OpenFolder…** command to navigate to the

/home/developer/Documents/code/Lab

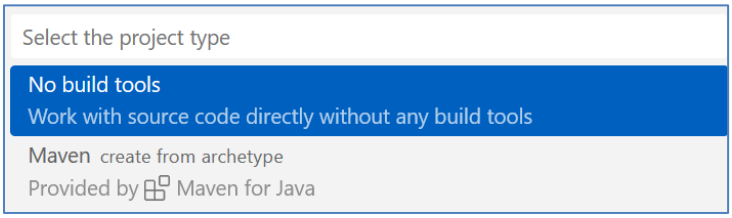
directory and click "**OK**".

8. The EXPLORER section of your screen should now look something like this:



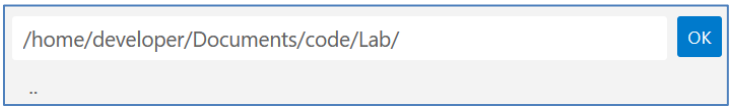
Create the new Java Project

9. Point your mouse in the **empty space** portion of the EXPLORER section of your screen, **right-click** on the empty space, and then **left-click** on the "Paste" button.
10. Select **New Java Project…**.
11. Observe the display:



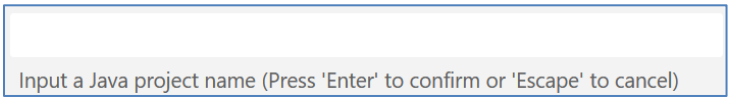
and select **No Build Tools**.

12. Observe the display:

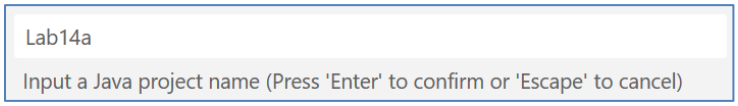


and click **OK**.

13. Observe the display

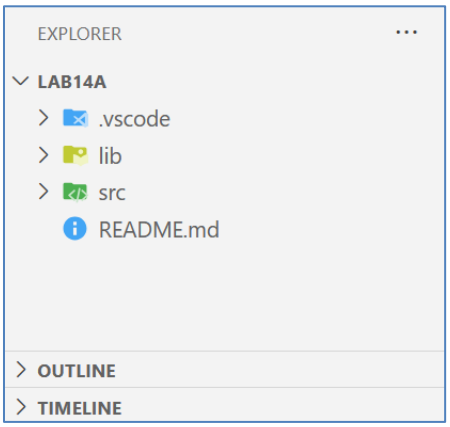


14. Type the project name into this field



and type the ENTER key.

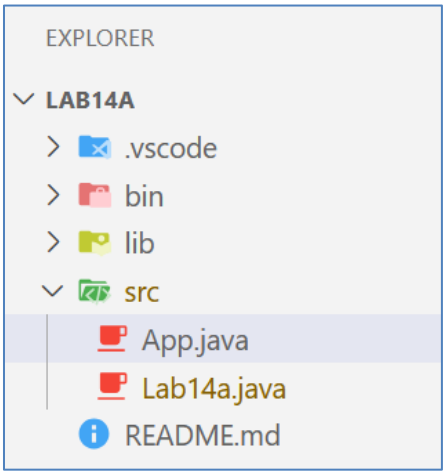
15. Now the EXPLORER section of the screen looks like



Right-click on the 

subdirectory and use the mouse-wheel to find and select the **Upload…** menu.

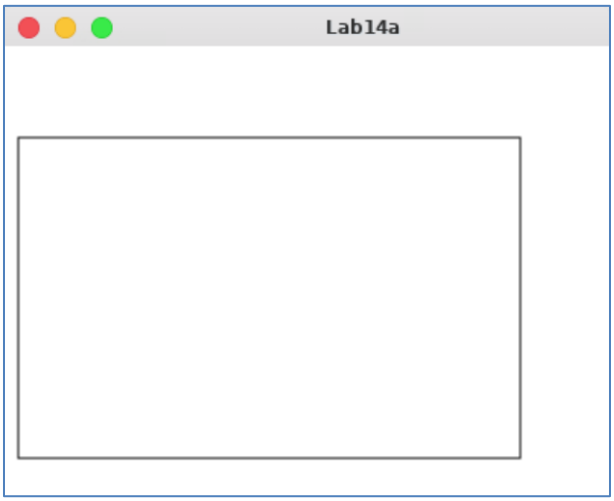
16. Observe the **File Upload** dialog box, and navigate to the folder (on your PC) where you saved the **Lab14a.java** file. Select the **Lab14a.java** file and click **Open**.
17. Now, the EXPLORER section of your screen will look something like this:

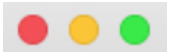


18. Before making any changes, try to **run the code** by clicking the  button near the upper-right corner of the screen. The program will run, but no **JavaFX** image will display yet. This is because you need to open a **Grader Than Desktop** *in addition* to the **Grader Than IDE** screen.
19. One way to open a **Grader Than Desktop** is to open another (simultaneous) **instance of your browser** and connect to **Grader Than** again, using the second browser. The new **Grader Than** display will include an image like:



20. The dark blue button on the right is actually <u>three</u> buttons. Click on the middle portion  , which selects the "**Open a desktop to access Apps**" operation.
21. Now you have two browsers running on your computer: one displays the source code, and the other displays the **JavaFX** image:



22. The three colored dots  at the upper-left corner of the image window let the user control some behavior of the display:
    - Clicking on the green dot  changes the image to "full screen" mode. Click on the green dot again to restore the normal display.
    - Clicking on the orange dot  minimizes the display, but does not stop the program.
    - Clicking on the red dot  causes the program to exit.

23. If you right-click on the title bar (at the top) of this window, then another menu appears:



You can experiment with this menu to understand what it does.

Required Program Enhancement

1. Add code to draw a second rectangle (at a slightly different location).

   HINT: Define the location of the new rectangle in terms of **`cornerX`** and **`cornerY`** in the starter program. Do not change the **`width`** and **`height`** values for the second rectangle.

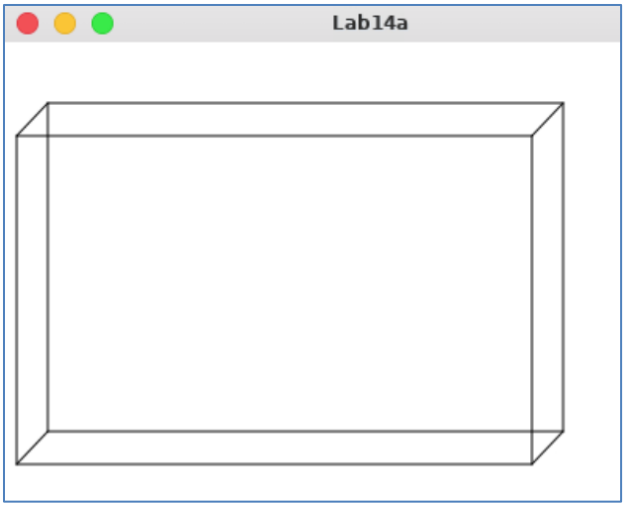
2. Test the program to observe two rectangles at the approximate desired positions.
3. Draw lines to connect the corresponding corners of the two rectangles.

   HINT: Calculate the **X** and **Y** values for all rectangle corners in terms of the **`cornerX`** and **`cornerY`** variables provided in the starter code. For example, the **X,Y** coordinates of the four corners of the **first** rectangle can be expressed as:

   ```
   cornerX, cornerY                    (upper-left)
   cornerX + width, cornerY            (upper-right)
   cornerX, cornerY + height           (lower-left)
   cornerX + width, cornerY +height    (lower-right)
   ```

   If you are having difficulty picturing this in your mind, use a **pencil and paper** to draw the first rectangle. Label the four corners. (Remember that the **Java** coordinate system has the Y-axis pointing *down*.)
   If you choose descriptive variable names for each value then drawing the lines may be easier.

The final display should look something like the following:



As always, submit the source code file to ***Moodle***, and demonstrate your solution during class.

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